

Selected-run evidence status: prepared

Geometry

Undeformed geometry

Exterior shape from the selected mesh



Mesh-derived exterior • source geometry unchanged

Undeformed geometry from selected mesh. Exterior shape derived from the selected run's mesh; this is not a CAD render.

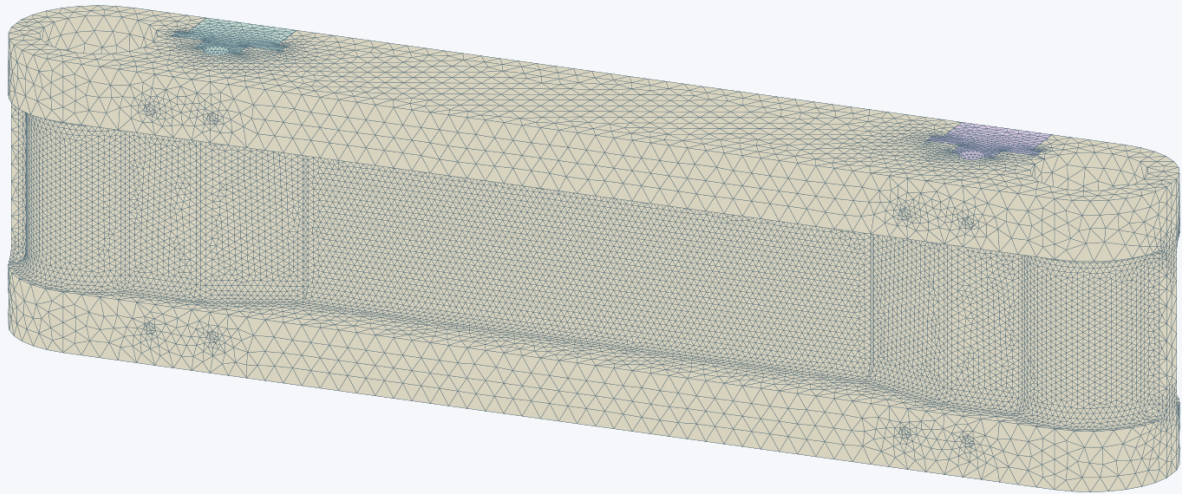
Mesh

Mesh overview and local detail

Mesh nodes	76873	
Tetrahedral elements	321831	
Meshed parts	3	
Component target sizes	5 / 3	mm
Minimum mean-ratio quality	0.0742	0-1

Mesh overview

Undeformed exterior mesh

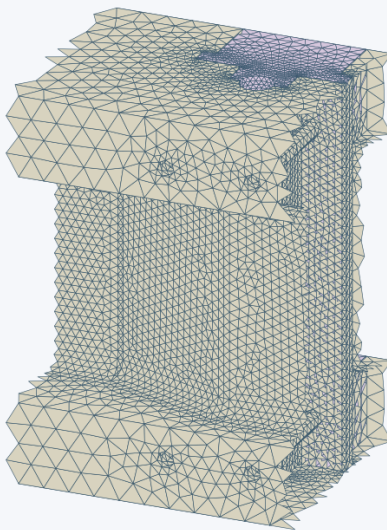


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Local mesh detail

Local mesh detail



Exterior mesh faces • undeformed geometry

Local mesh detail. Local view of the selected mesh; no local refinement claim is made.

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	8	ms
Foam depth	203.2	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.05	ms
Startup mass-scaling DT2MS	-1.5e-07	s
Termination time-step ratio	0.01	dimensionless
Printed parts mass	0.5627708	kg
Hardware mass	0.03400217	kg
Motor mass	0.16	kg
Moving mass	0.756773	kg
Initial kinetic energy	945.9662	J

Recorded material, contact and boundary assumptions

- Unchanged prior clamshell fixture and synthetic 54.46 kg/m³ MAT057 foam; not the proposed 30 kg/m³ product.
- Three independent printed solids; body-to-closure frictional contact plus eight ideal bolt collars; no slip/preload/failure capacity.
- Eight CAD generic steel bolt/nut envelope masses distributed on the collar nodes.
- Two 80 g motor masses distributed on the original motor bores; no casing stiffness.
- PA12-CF10 isotropic linear elastic screen; not printed anisotropic strength qualification.
- Ideal transverse cable-bore guidance leaves axial Z free; no cable compliance, tension or clearance contact.
- Two stationary aluminum spreaders, 0.1 mm initial gap; fixed foam rear Z, free foam sides.

- No carrier, motor thrust, element failure, fracture or damage.

Solve

Recorded run status

prepared

Separate solve charts

Contact force resultant

No extracted history for this result in the selected run.

Contact force resultant. No extracted source available.

Mass-weighted forward velocity

No extracted history for this result in the selected run.

Mass-weighted forward velocity. No extracted source available.

Moving assembly travel

No extracted history for this result in the selected run.

Moving assembly travel. No extracted source available.

Foam compression

No extracted history for this result in the selected run.

Foam compression. No extracted source available.

Regional P99 von Mises stress

No extracted history for this result in the selected run.

Regional P99 von Mises stress. No extracted source available.

Global energy

No extracted history for this result in the selected run.

Global energy. No extracted source available.

Results

Run 036 - impact evidence

Evidence status: prepared. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

Geometry and model setup

Recorded model item	Value
kind	impact
assembly kind	local_closures
cad root	revised
printed kg	0.562771
hardware kg	0.0340022
motors kg	0.16
moving kg	0.756773
initial ke J	945.966
spreader each kg	0.135871

Solved mesh

Recorded mesh source: stem_5p5_5mm. Mesh and timestep convergence are not established by these histories.

Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

Measured result	Value	Unit
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Regional stress and energy

Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

Numerical checks and qualification

No selected-run numerical check summary is available.

No extracted results are available for this run yet.

Images, animations and source evidence

- model.json
- setup_audit.json

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.